Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.2642377632041777, median 0.2004443087197021, std: 0.24673804097225918
Gyroscope error (imu0): mean 0.3123942715088874, median 0.1783297871274419, std: 0.5450654581507828
Accelerometer error (imu0): mean 0.4857513550548161, median 0.3194829041905828, std: 0.7300714717262565

Residuals

Reprojection error (cam0) [px]: mean 0.2642377632041777, median 0.2004443087197021, std: 0.24673804097225918
Gyroscope error (imu0) [rad/s]: mean 0.00616731050463646, median 0.003520599670821324, std: 0.010760722050155045
Accelerometer error (imu0) [m/s^2]: mean 0.09429597787975733, median 0.062019287343246146, std: 0.14172436707000474

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.22393449 -0.95189124 0.20918033 -0.03649457]
 [0.38494661 -0.28356735 -0.87829702 0.01953115]
 [0.89535996 -0.11615773 0.42992782 -0.01729193]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.22393449 0.38494661 0.89535996 -0.00020834]
 [-0.95189124 -0.28356735 -0.11615773 -0.03120906]
 [0.20918033 -0.87829702 0.42992782 0.03222238]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.00739080791303851

Gravity vector in target coords: [m/s^2]
[-0.19920342 -9.40973356 2.75420682]

Calibration configuration
==========

cam0

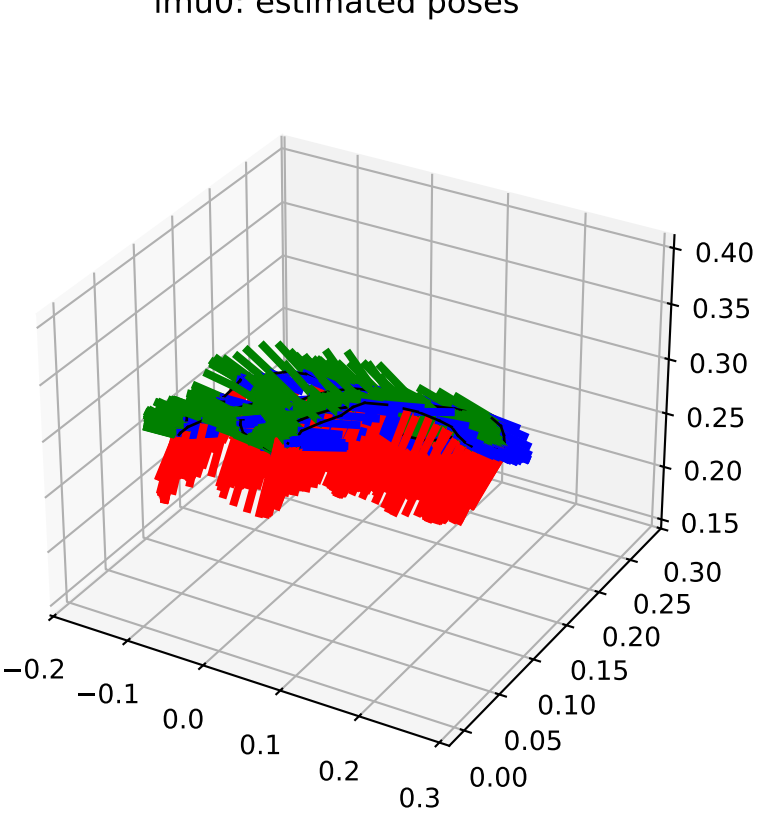
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.017733333333333334 [m]
    Spacing 0.00532 [m]

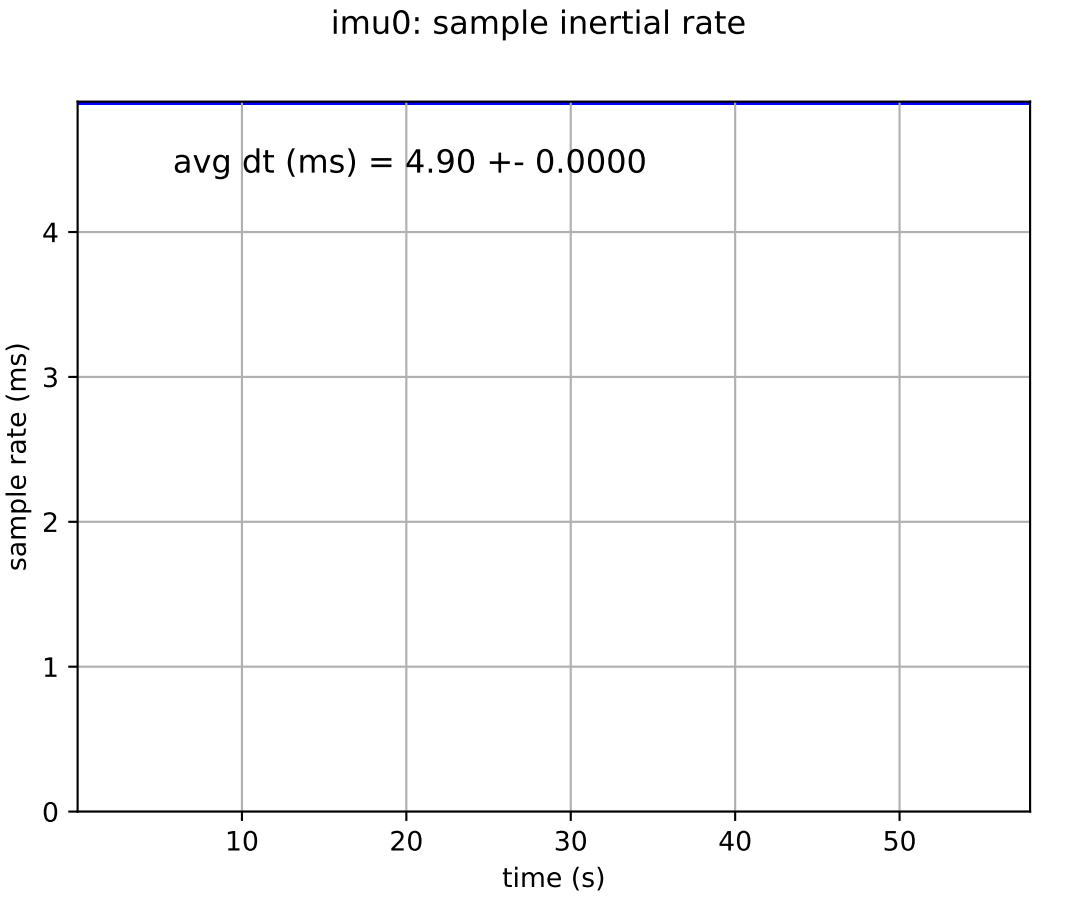
IMU configuration
==========

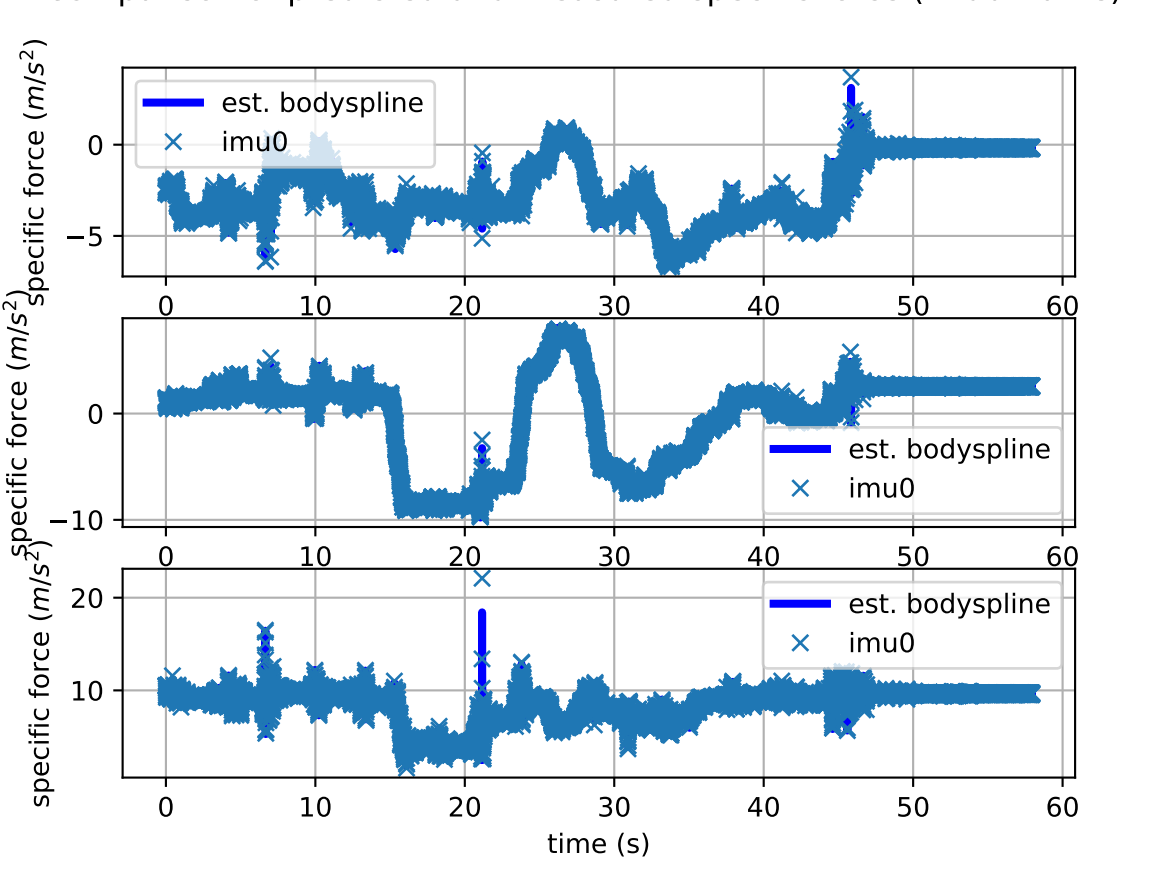
IMU0:

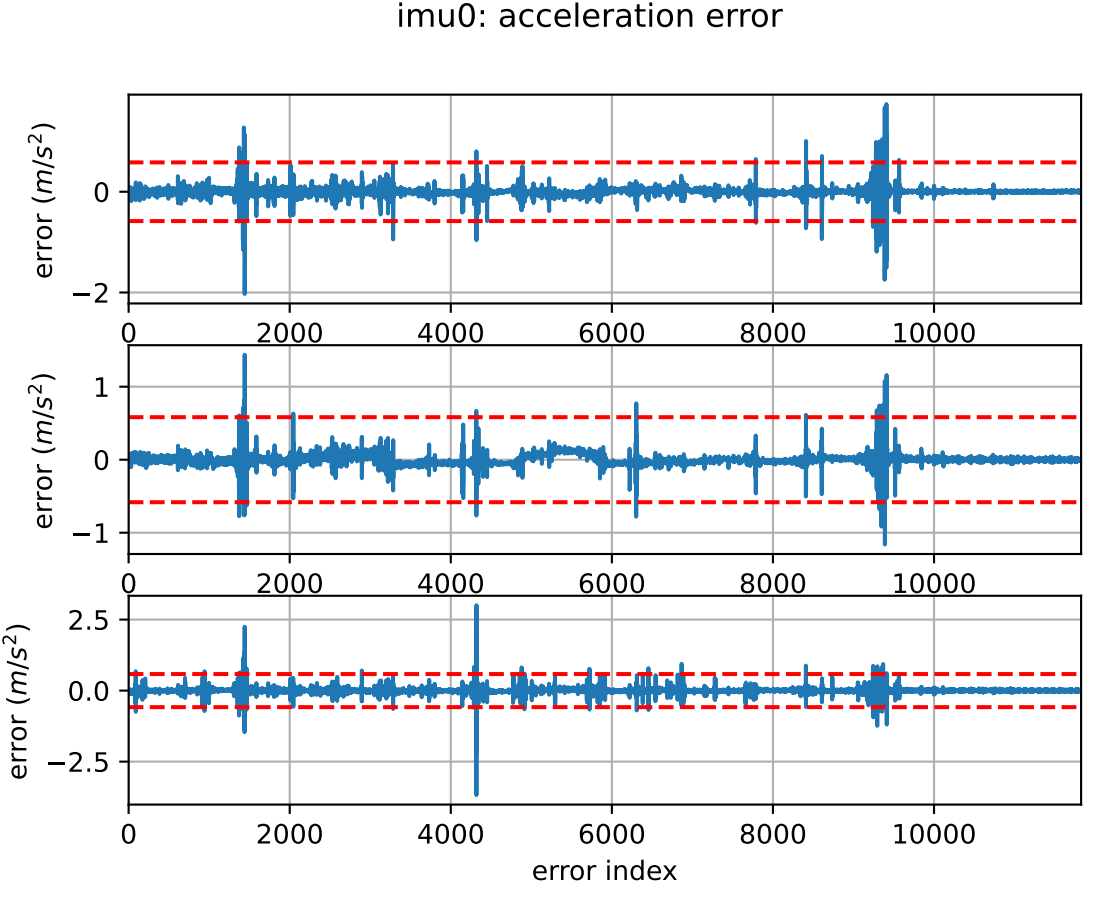
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.013588383144412769
    Noise density (discrete): 0.19412396259628797
    Random walk: 0.0033511810792688133
  Gyroscope:
    Noise density: 0.0013819151847161266
    Noise density (discrete): 0.019742072973514827
    Random walk: 7.385130430490958e-05

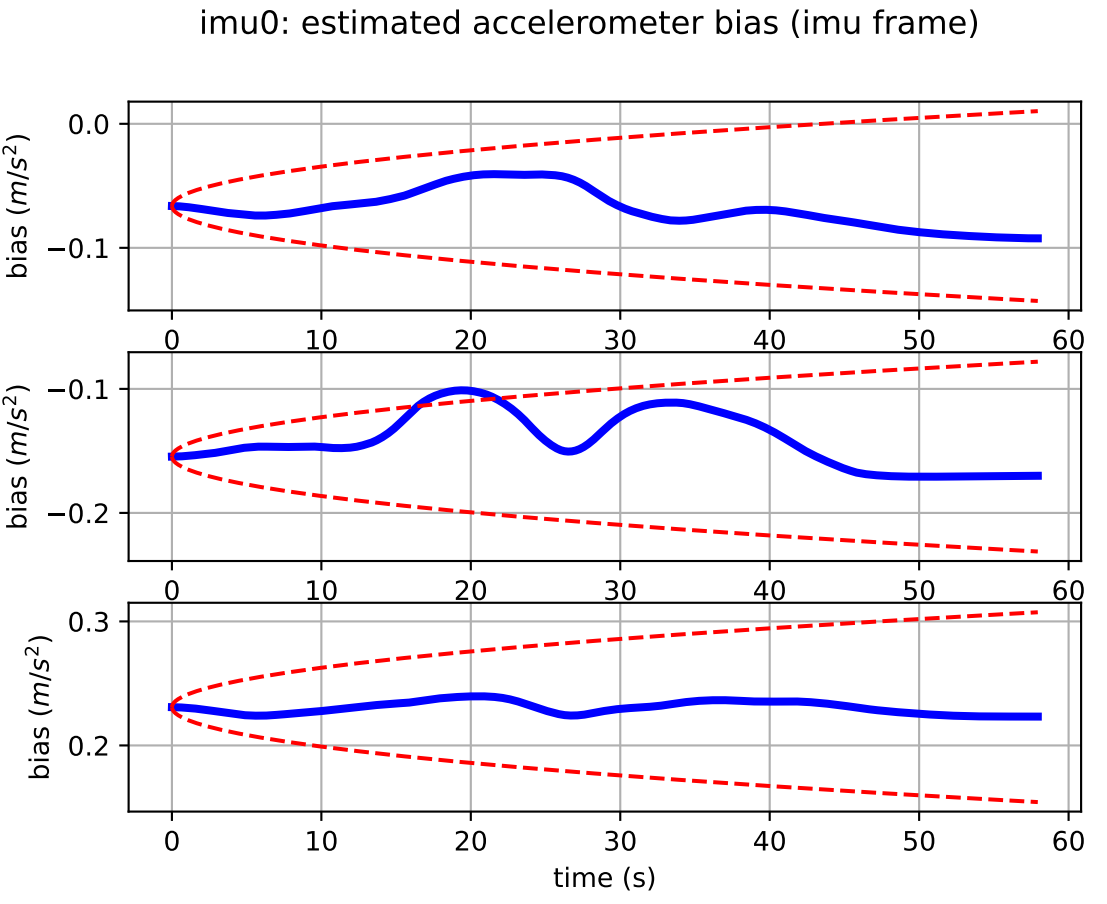
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

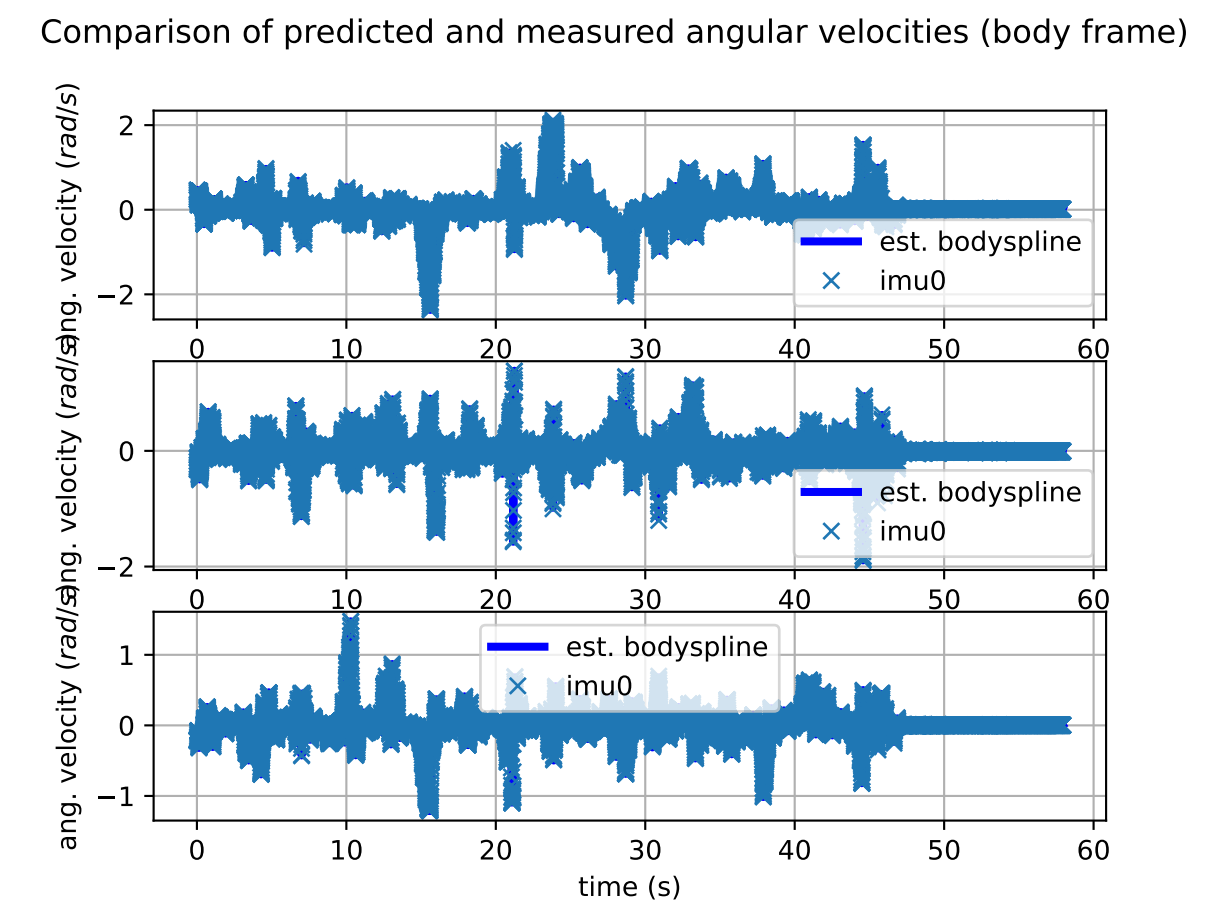

imu0: estimated poses
0.40
0.35
0.30
0.25
0.20
0.15
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.2
−0.1
0.0
0.1
0.2
0.3

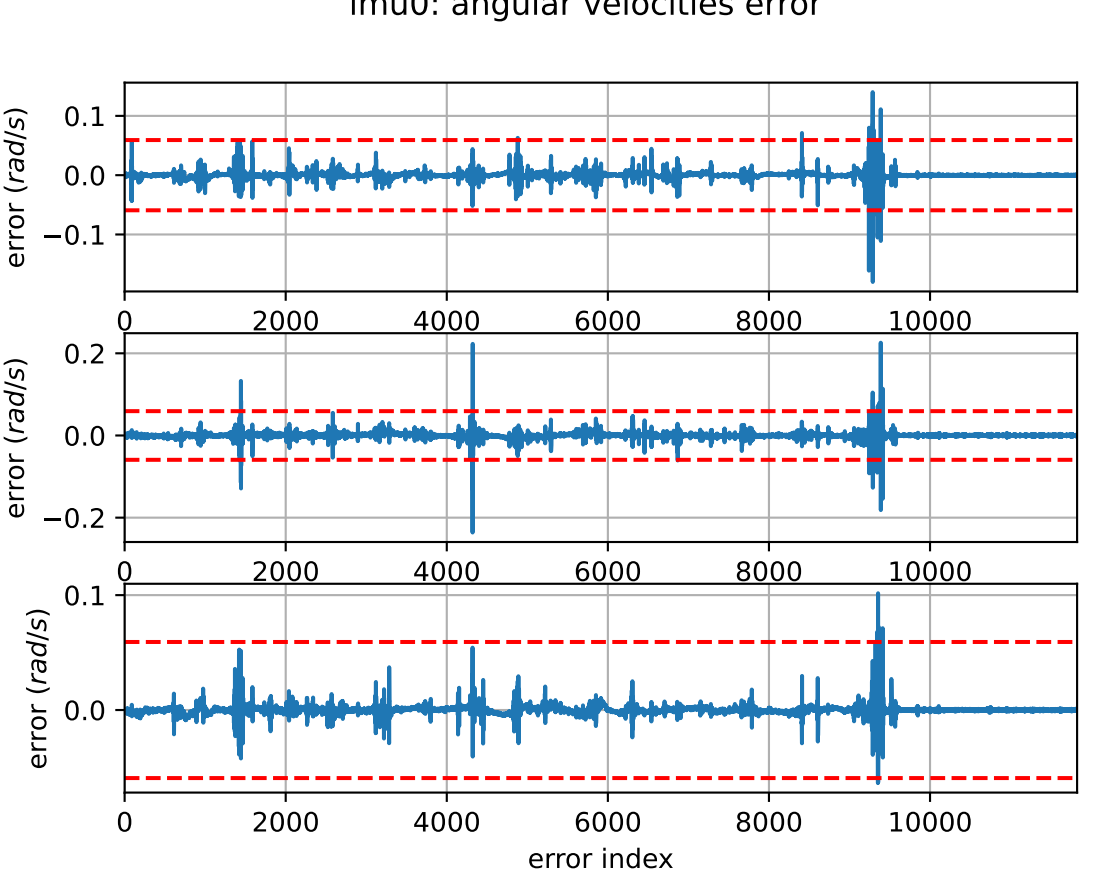

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0000
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

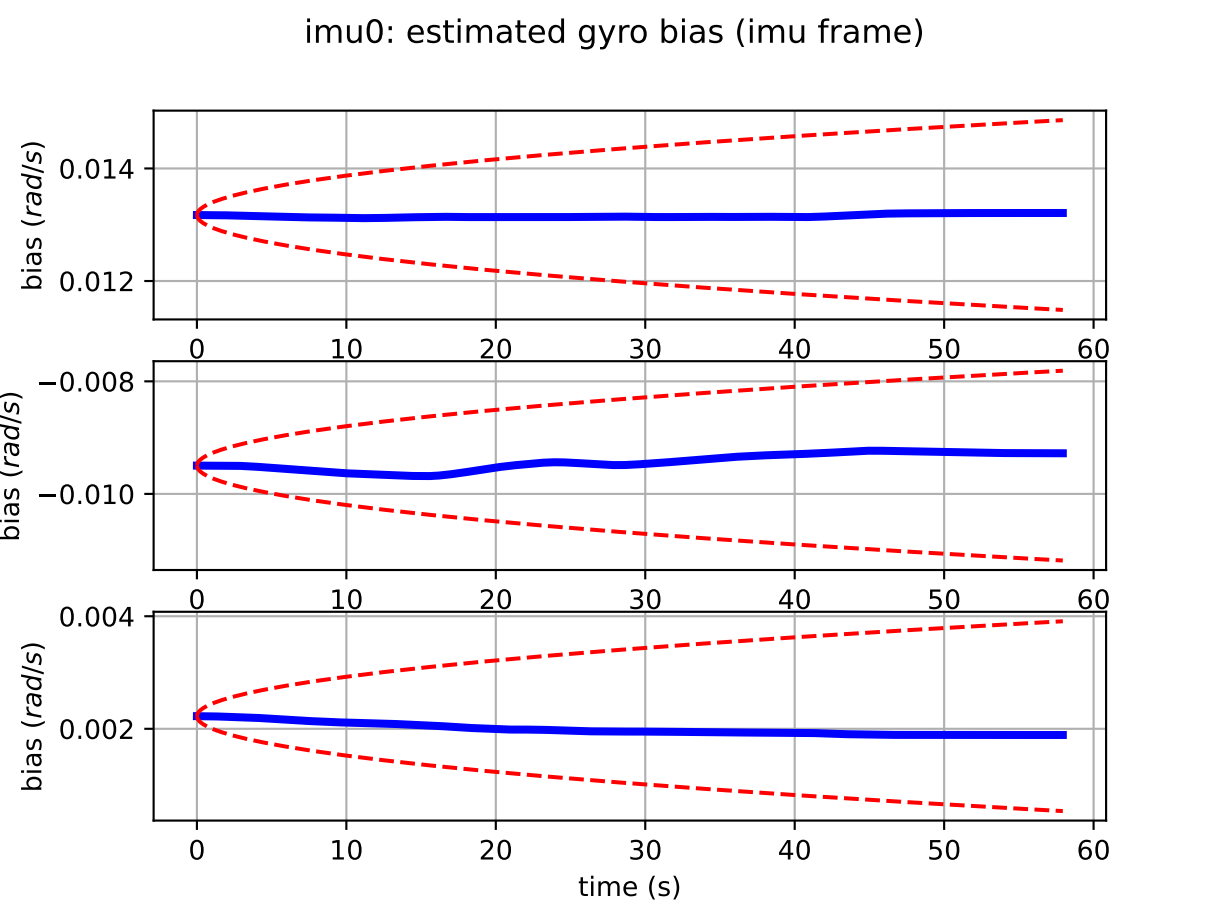

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
0
−5
0
−10
20
10
0
10
20
30
40
50
60
time (s)

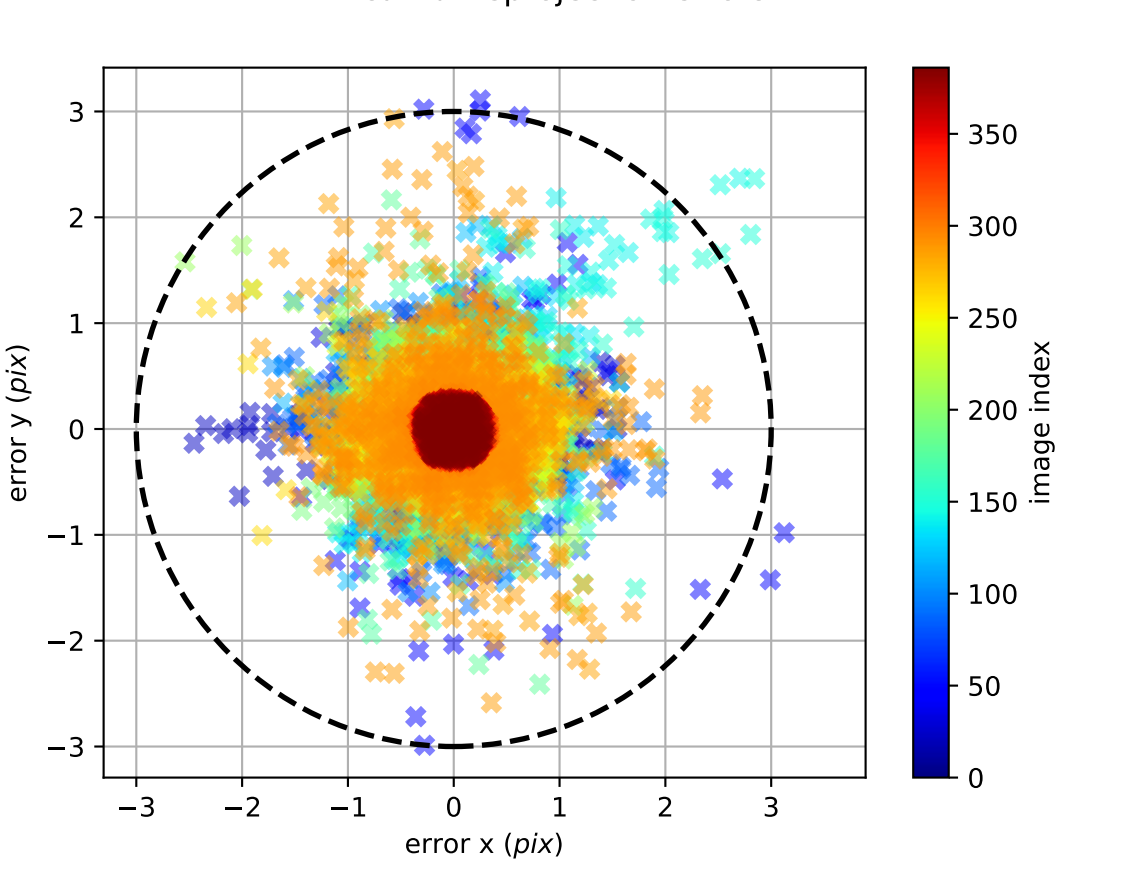

imu0: acceleration error
error (m/s^2)
0
−2
0
2000
4000
6000
8000
10000
error (m/s^2)
1
0
−1
0
2000
4000
6000
8000
10000
error (m/s^2)
2.5
0.0
−2.5
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.0
−0.1
0
10
20
30
40
50
60
bias (m/s^2)
−0.1
−0.2
0
10
20
30
40
50
60
bias (m/s^2)
0.3
0.2
0
10
20
30
40
50
60
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
time (s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
2000
4000
6000
8000
10000
error (rad/s)
0.2
0.0
−0.2
0
2000
4000
6000
8000
10000
error (rad/s)
0.1
0.0
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.014
0.012
0
10
20
30
40
50
60
bias (rad/s)
−0.008
−0.010
bias (rad/s)
0.004
0.002
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−3
−2
−1
0
1
2
3
0
50
100
150
200
250
300
350